

Title: Enterprise ISP Network Design and Implementation Using OSPF, eBGP, pfSense Firewall, and Ubuntu Data Center

Abstract:

This project implements a secure enterprise ISP network using Cisco routers, OSPF, eBGP, pfSense Firewall, and an Ubuntu Server. The network provides secure routing, firewall protection, NAT, and centralized services including Apache, DNS, SSH, Syslog and FTP.

Introduction:

The objective of this project is to simulate a real enterprise network where multiple customer networks communicate through an ISP while being protected by a firewall and supported by centralized server services.

Objectives:

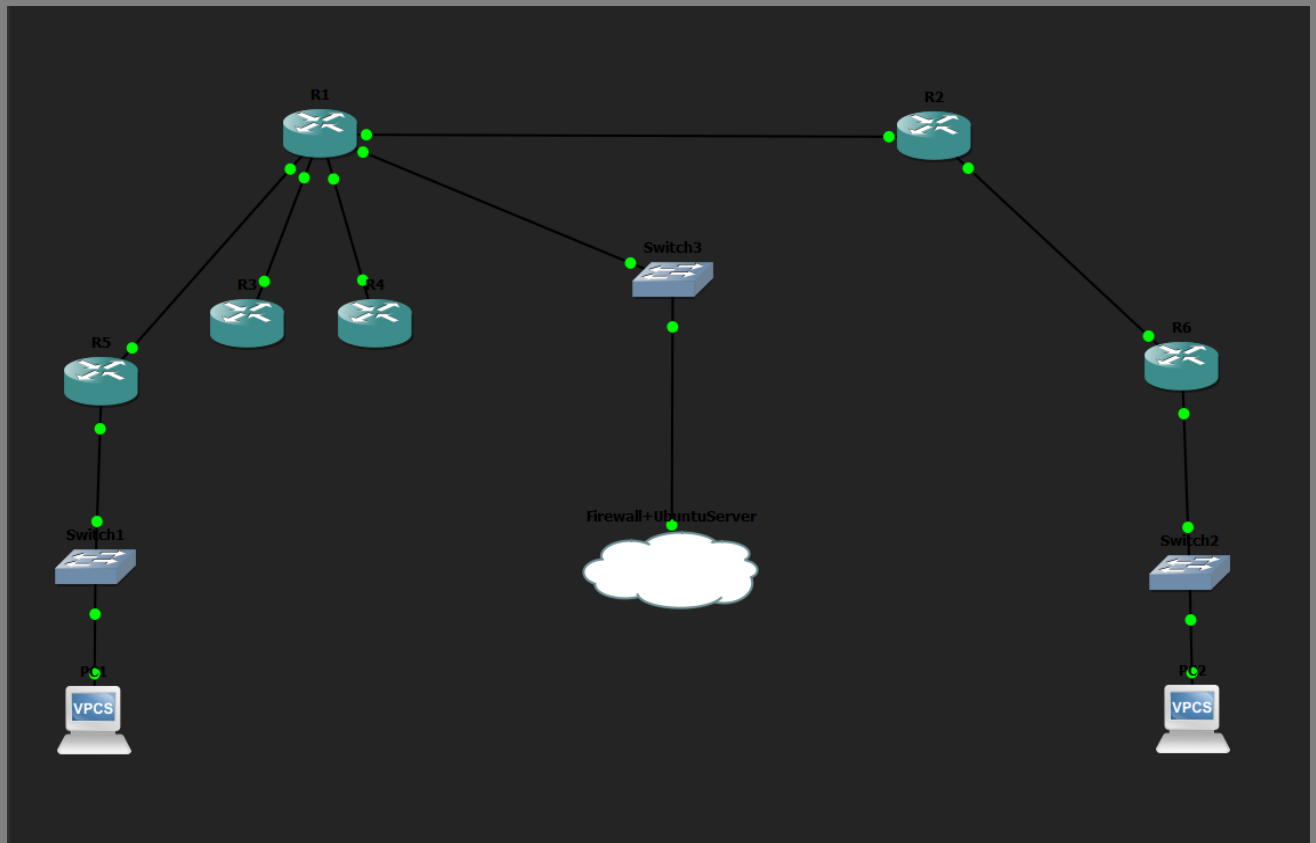
- Design an enterprise ISP topology.
- Configure OSPF and eBGP.
- Implement Prefix Lists, Local Preference and ACLs.
- Configure pfSense Firewall and NAT.
- Deploy Ubuntu Server services.
- Verify complete connectivity.

Hardware and Software Requirements:

Hardware: PC, Laptop, Ethernet Cable.

Software: GNS3, VMware Workstation, Cisco IOS, pfSense, Ubuntu Server, Apache2, BIND9, OpenSSH, rsyslog, vsftpd.

Network Topology:



IP Addressing:

Device	Interface	IP
R1	E1/3	192.168.50.10
pfSense	OPT1	192.168.50.1
Ubuntu	eth0	192.168.50.21
Windows PC	Ethernet	192.168.50.11
Laptop	Ethernet	192.168.50.12

OSPF Configuration:

OSPF neighbor

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	FULL/DR	00:00:33	10.0.0.14	Ethernet1/2
4.4.4.4	1	FULL/DR	00:00:32	10.0.0.10	Ethernet1/1
3.3.3.3	1	FULL/DR	00:00:33	10.0.0.6	Ethernet1/0
2.2.2.2	1	FULL/DR	00:00:34	10.0.0.2	FastEthernet0/0

```
R1#
```

OSPF Route:

```
R1#show ip route ospf
```

```
    2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.0.0.2, 01:49:01, FastEthernet0/0
    6.0.0.0/32 is subnetted, 1 subnets
O       6.6.6.6 [110/12] via 10.0.0.2, 01:49:01, FastEthernet0/0
O    192.168.20.0/24 [110/21] via 10.0.0.2, 01:49:01, FastEthernet0/0
    10.0.0.0/30 is subnetted, 5 subnets
O       10.0.0.16 [110/11] via 10.0.0.2, 01:49:01, FastEthernet0/0
```

```
R1#
```

Show IP Protocols:

```
R1#show ip protocols
```

```
Routing Protocol is "ospf 1"
```

```
  Outgoing update filter list for all interfaces is not set
```

```
  Incoming update filter list for all interfaces is not set
```

```
  Router ID 1.1.1.1
```

```
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
  Maximum path: 4
```

```
  Routing for Networks:
```

```
    1.1.1.1 0.0.0.0 area 0
```

```
    10.0.0.0 0.0.0.255 area 0
```

```
    172.16.100.0 0.0.0.255 area 0
```

```
    192.168.10.0 0.0.0.255 area 0
```

```
    192.168.50.0 0.0.0.255 area 0
```

```
Reference bandwidth unit is 100 mbps
```

```
Routing Information Sources:
```

Gateway	Distance	Last Update
---------	----------	-------------

2.2.2.2	110	01:50:00
---------	-----	----------

6.6.6.6	110	01:50:00
---------	-----	----------

5.5.5.5	110	01:50:10
---------	-----	----------

```
Distance: (default is 110)
```

```
Routing Protocol is "bgp 65001"
```

```
  Outgoing update filter list for all interfaces is not set
```

```
  Incoming update filter list for all interfaces is not set
```

```
  IGP synchronization is disabled
```

```
  Automatic route summarization is disabled
```

```
Neighbor(s):
```

Address	FiltIn	FiltOut	DistIn	DistOut	Weight	RouteMap
10.0.0.14						CUSTOMER-A

```
Maximum path: 1
```

```
Routing Information Sources:
```

Gateway	Distance	Last Update
---------	----------	-------------

10.0.0.14	20	01:50:02
-----------	----	----------

```
Distance: external 20 internal 200 local 200
```

eBGP Configuration:

BGP Summary

```
R1#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 65001
BGP table version is 5, main routing table version 5
4 network entries using 528 bytes of memory
4 path entries using 208 bytes of memory
4/2 BGP path/bestpath attribute entries using 672 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1464 total bytes of memory
BGP activity 4/0 prefixes, 4/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ  OutQ Up/Down  State/PfxRcd
10.0.0.14      4      65010    114     114       5    0    0 01:52:19      2
```

show ip bgp

```
R1#show ip bgp
BGP table version is 5, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop        Metric LocPrf Weight Path
*> 0.0.0.0           0.0.0.0          0         32768 i
*> 5.5.5.5/32        10.0.0.14         0         200      0 65010 i
*> 192.168.10.0      10.0.0.14         0         200      0 65010 i
*> 192.168.50.0      0.0.0.0          0         32768 i
```

show ip route bgp

```
R1#show ip route BGP
      5.0.0.0/32 is subnetted, 1 subnets
B       5.5.5.5 [20/0] via 10.0.0.14, 01:52:55
B      192.168.10.0/24 [20/0] via 10.0.0.14, 01:52:55
R1#
```

Routing Verification:

Show IP Route

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    1.0.0.0/32 is subnetted, 1 subnets
C      1.1.1.1 is directly connected, Loopback0
    2.0.0.0/32 is subnetted, 1 subnets
O      2.2.2.2 [110/2] via 10.0.0.2, 01:53:52, FastEthernet0/0
    5.0.0.0/32 is subnetted, 1 subnets
B      5.5.5.5 [20/0] via 10.0.0.14, 01:53:50
B     192.168.10.0/24 [20/0] via 10.0.0.14, 01:53:50
    6.0.0.0/32 is subnetted, 1 subnets
O      6.6.6.6 [110/12] via 10.0.0.2, 01:53:52, FastEthernet0/0
   172.16.0.0/24 is subnetted, 1 subnets
C     172.16.100.0 is directly connected, Ethernet2/0
O     192.168.20.0/24 [110/21] via 10.0.0.2, 01:53:52, FastEthernet0/0
   10.0.0.0/30 is subnetted, 5 subnets
C      10.0.0.8 is directly connected, Ethernet1/1
C      10.0.0.12 is directly connected, Ethernet1/2
C      10.0.0.0 is directly connected, FastEthernet0/0
C      10.0.0.4 is directly connected, Ethernet1/0
O      10.0.0.16 [110/11] via 10.0.0.2, 01:53:57, FastEthernet0/0
C     192.168.50.0/24 is directly connected, Ethernet1/3
S*    0.0.0.0/0 is directly connected, Null0
R1#
```

Show IP int Brief:

```
R1#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.1        YES NVRAM    up          up
Ethernet1/0               10.0.0.5        YES NVRAM    up          up
Ethernet1/1               10.0.0.9        YES NVRAM    up          up
Ethernet1/2               10.0.0.13       YES NVRAM    up          up
Ethernet1/3               192.168.50.10   YES NVRAM    up          up
Ethernet2/0               172.16.100.1    YES NVRAM    up          up
Ethernet2/1               unassigned      YES NVRAM    administratively down down
Ethernet2/2               unassigned      YES NVRAM    administratively down down
Ethernet2/3               unassigned      YES NVRAM    administratively down down
Ethernet3/0               unassigned      YES NVRAM    administratively down down
Ethernet3/1               unassigned      YES NVRAM    administratively down down
Ethernet3/2               unassigned      YES NVRAM    administratively down down
Ethernet3/3               unassigned      YES NVRAM    administratively down down
Loopback0                 1.1.1.1         YES NVRAM    up          up
R1#
```

ACL Verification:

Show access-list

```
R1#  
R1#show access-list  
Extended IP access list BLOCK-CUSTOMERS  
 10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255  
 20 permit ip any any  
R1#
```

Prefix List Verification:

```
R1#show ip prefix-list
ip prefix-list COMPANY-A: 2 entries
  seq 10 permit 192.168.10.0/24
  seq 20 permit 5.5.5.5/32
R1#
```

pfSense Verification:

Interfaces (WAN/LAN/OPT1):

Interfaces ⚙️ - ✕			
 WAN	↑	1000baseT <full-duplex>	192.168.10.6
 LAN	↑	1000baseT <full-duplex>	192.168.75.2
 OPT1	↑	1000baseT <full-duplex>	192.168.50.1

Firewall > Rules

Firewall / Rules / OPT1

Floating

WAN

LAN

OPT1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	6/927.38 MiB	IPv4 *	*	*	*	*	*	none			
☐	0/0 B	IPv4 TCP	192.168.10.10/24	*	*	22 (SSH)	*	none		Admin SSH	
☐	0/0 B	IPv4 TCP	*	*	*	53 (DNS)	*	none		Allow DNS	
☐	0/0 B	IPv4 TCP	*	*	*	443 (HTTPS)	*	none		Allow HTTPS	
☐	0/0 B	IPv4 TCP	*	*	*	80 (HTTP)	*	none		HTTP(80)	

Add

Add

Delete

Toggle

Copy

Save

Separate

Routing Gateways:

System / Routing / Gateways

Gateways

Static Routes

Gateway Groups

Gateways							
	Name	Default	Interface	Gateway	Monitor IP	Description	Actions
☐	WAN_DHCP6		WAN	fe80::aa63:7dff:fe82:cad7%em0	fe80::aa63:7dff:fe82:cad7%em0	Interface WAN_DHCP6 Gateway	
☐	WAN_DHCP		WAN	192.168.10.1	192.168.10.1	Interface WAN_DHCP Gateway	
☐	WANGW	Default (IPv4)	WAN	192.168.10.1	192.168.10.1	Interface wan Gateway	
☐	LAN_DHCP		LAN	192.168.75.1	192.168.75.1	Interface LAN_DHCP Gateway	
☐	LANGW		LAN	192.168.26.1	192.168.26.1	Interface lan Gateway	
☐	LAN_DHCP6		LAN	dynamic	dynamic	Interface LAN_DHCP6 Gateway	
☐	LANGW_2		LAN	192.168.75.1	192.168.75.1	Interface lan Gateway	
☐	R1_GW		OPT1	192.168.50.10	192.168.50.10		

Save

Add

Static Routes:

System / Routing / Static Routes

Gateways

Static Routes

Gateway Groups

Static Routes					
	Network	Gateway	Interface	Description	Actions
☒	192.168.10.0/24	R1_GW - 192.168.50.10	OPT1		
☒	192.168.20.0/24	R1_GW - 192.168.50.10	OPT1		
☒	10.0.0.16/30	R1_GW - 192.168.50.10	OPT1		
☒	10.0.0.12/30	R1_GW - 192.168.50.10	OPT1		

+

Add

i

Pinging pfSense To Company A:

Ping

Hostname

192.168.10.1

IP Protocol

IPv4

Source address

Automatically selected (default)

Select source address for the ping.

Maximum number of pings

3

Select the maximum number of pings.

Seconds between pings

1

Select the number of seconds to wait between pings.

Ping

Results

PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: icmp_seq=0 ttl=64 time=3.288 ms
64 bytes from 192.168.10.1: icmp_seq=1 ttl=64 time=4.608 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=64 time=2.228 ms

--- 192.168.10.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.228/3.372/4.608/0.974 ms

Pinging pfSense To Company B:

Ping	
Hostname	192.168.20.1
IP Protocol	IPv4
Source address	Automatically selected (default) Select source address for the ping.
Maximum number of pings	3 Select the maximum number of pings.
Seconds between pings	1 Select the number of seconds to wait between pings.

 Ping

Results	
PING 192.168.20.1 (192.168.20.1): 56 data bytes 64 bytes from 192.168.20.1: icmp_seq=0 ttl=253 time=65.588 ms 64 bytes from 192.168.20.1: icmp_seq=1 ttl=253 time=70.531 ms 64 bytes from 192.168.20.1: icmp_seq=2 ttl=253 time=68.180 ms --- 192.168.20.1 ping statistics --- 3 packets transmitted, 3 packets received, 0.0% packet loss round-trip min/avg/max/stddev = 65.588/68.100/70.531/2.019 ms	

Ubuntu Server Verification:

ip addr

```
haider@linux-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:31:20:1b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.50.21/24 brd 192.168.50.255 scope global ens33
        valid_lft forever preferred_lft forever
haider@linux-server:~$
```

ip route

```
haider@linux-server:~$ ip route
default via 192.168.50.1 dev ens33 proto static
192.168.50.0/24 dev ens33 proto kernel scope link src 192.168.50.21
haider@linux-server:~$
```


hostname -I

```
haider@linux-server:~$ hostname -I
192.168.50.21
haider@linux-server:~$ _
```

Apache Verification:

systemctl status apache2

```
haider@linux-server:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-02 06:32:33 UTC; 2h 15min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1327 (apache2)
    Tasks: 55 (limit: 6540)
   Memory: 8.1M (peak: 8.5M)
      CPU: 577ms
   CGroup: /system.slice/apache2.service
           └─1327 /usr/sbin/apache2 -k start
             └─1335 /usr/sbin/apache2 -k start
               └─1336 /usr/sbin/apache2 -k start

Jul 02 06:32:32 linux-server systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 02 06:32:32 linux-server apachectl[1259]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the
Jul 02 06:32:33 linux-server systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-16/16 (END)
```

Open <http://192.168.50.21>

Not secure192.168.50.21

☆🔖⬇️👤School

ISP Data Center

Ubuntu Web Server

Services Running

Apache Web Server

DNS Server

FTP Server

SSH Server

SYSlog Server

Server INFO

Server IP: 192.168.50.21

Firewall: PFsense

Routing: OSPF + eBGP

Developed By

Ali Haider

DNS Verification:

systemctl status bind9

```
haider@linux-server:~$ systemctl status bind9
• named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-02 06:32:32 UTC; 2h 16min ago
     Docs: man:named(8)
    Main PID: 868 (named)
      Status: "running"
        Tasks: 14 (limit: 6540)
      Memory: 28.7M (peak: 29.4M)
         CPU: 525ms
    CGroup: /system.slice/named.service
            └─868 /usr/sbin/named -f -u bind

Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './NS/IN': 2001:500:2f::f#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './DNSKEY/IN': 2001:7fe::53#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './NS/IN': 2001:7fe::53#53
Jul 02 07:32:42 linux-server named[868]: network unreachable resolving './DNSKEY/IN': 2001:500:1::53#53
Jul 02 07:32:42 linux-server named[868]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer complete)
Jul 02 07:32:42 linux-server named[868]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance timer complete)
haider@linux-server:~$ _
```

nslookup www.isp.local 127.0.0.1

```
haider@linux-server:~$ nslookup www.isp.local 127.0.0.1
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   www.isp.local
Address: 192.168.50.21

haider@linux-server:~$ _
```

SSH Verification:

systemctl status ssh

```
haider@linux-server:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-02 06:32:42 UTC; 2h 18min ago
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1782 (sshd)
     Tasks: 1 (limit: 6540)
    Memory: 2.1M (peak: 2.2M)
       CPU: 26ms
   CGroup: /system.slice/ssh.service
           └─1782 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 02 06:32:42 linux-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 02 06:32:42 linux-server sshd[1782]: Server listening on 0.0.0.0 port 22.
Jul 02 06:32:42 linux-server sshd[1782]: Server listening on :: port 22.
Jul 02 06:32:42 linux-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
haider@linux-server:~$ _
```

ssh haider@192.168.50.21

```
haider@linux-server: ~  
Microsoft Windows [Version 10.0.19045.6456]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\aliha>ssh haider@192.168.50.21  
haider@192.168.50.21's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Thu Jul  2 08:51:20 AM UTC 2026  
  
System load:  0.0                Processes:            256  
Usage of /:   32.0% of 11.21GB    Users logged in:     1  
Memory usage: 7%                IPv4 address for ens33: 192.168.50.21  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
139 updates can be applied immediately.  
111 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Thu Jul  2 08:51:21 2026 from 192.168.50.21  
haider@linux-server:~$
```

Verification:

```
haider@linux-server:~$ ifconfig  
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 192.168.50.21  netmask 255.255.255.0  broadcast 192.168.50.255  
    ether 00:0c:29:31:20:1b  txqueuelen 1000  (Ethernet)  
    RX packets 647515  bytes 962794854 (962.7 MB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 322181  bytes 21651386 (21.6 MB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536  
    inet 127.0.0.1  netmask 255.0.0.0  
    inet6 ::1  prefixlen 128  scopeid 0x10<host>  
    loop txqueuelen 1000  (Local Loopback)  
    RX packets 255  bytes 30634 (30.6 KB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 255  bytes 30634 (30.6 KB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
haider@linux-server:~$
```

Syslog Verification:

systemctl status rsyslog

```
haider@linux-server:~$ systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-02 06:32:31 UTC; 2h 20min ago
   TriggeredBy: ● syslog.socket
     Docs: man:rsyslogd(8)
           man:rsyslog.conf(5)
           https://www.rsyslog.com/doc/
   Main PID: 942 (rsyslogd)
     Tasks: 5 (limit: 6540)
    Memory: 3.8M (peak: 5.2M)
       CPU: 532ms
    CGroup: /system.slice/rsyslog.service
            └─942 /usr/sbin/rsyslogd -n -iNONE

Jul 02 06:32:30 linux-server systemd[1]: Starting rsyslog.service - System Logging Service...
Jul 02 06:32:31 linux-server rsyslogd[942]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd. [v8.2312.0]
Jul 02 06:32:31 linux-server rsyslogd[942]: rsyslogd's groupid changed to 104
Jul 02 06:32:31 linux-server rsyslogd[942]: rsyslogd's userid changed to 103
Jul 02 06:32:31 linux-server rsyslogd[942]: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="942" x-info="https://www.rsyslog.com"] start
Jul 02 06:32:31 linux-server systemd[1]: Started rsyslog.service - System Logging Service.
haider@linux-server:~$ _
```

Log Buffer:

```
Log Buffer (8192 bytes):

*Jul  2 11:44:28.035: % Error opening nvram:/ifIndex-table No such file or directory
*Jul  2 11:44:40.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface VoIP-Null0, changed state to up
*Jul  2 11:44:40.955: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Jul  2 11:44:40.963: %LINK-3-UPDOWN: Interface Ethernet1/0, changed state to up
*Jul  2 11:44:40.967: %LINK-3-UPDOWN: Interface Ethernet1/1, changed state to up
*Jul  2 11:44:40.971: %LINK-3-UPDOWN: Interface Ethernet1/2, changed state to up
*Jul  2 11:44:40.975: %LINK-3-UPDOWN: Interface Ethernet1/3, changed state to up
*Jul  2 11:44:40.979: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up
*Jul  2 11:44:40.987: %LINK-3-UPDOWN: Interface Ethernet2/1, changed state to up
*Jul  2 11:44:40.991: %LINK-3-UPDOWN: Interface Ethernet2/2, changed state to up
*Jul  2 11:44:40.995: %LINK-3-UPDOWN: Interface Ethernet2/3, changed state to up
*Jul  2 11:44:40.999: %LINK-3-UPDOWN: Interface Ethernet3/0, changed state to up
*Jul  2 11:44:41.007: %LINK-3-UPDOWN: Interface Ethernet3/1, changed state to up
*Jul  2 11:44:41.011: %LINK-3-UPDOWN: Interface Ethernet3/2, changed state to up
*Jul  2 11:44:41.011: %LINK-3-UPDOWN: Interface Ethernet3/3, changed state to up
*Jul  2 11:44:41.919: %SYS-5-CONFIG I: Configured from memory by console
*Jul  2 11:44:42.027: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
*Jul  2 11:44:42.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to up
*Jul  2 11:44:42.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/1, changed state to up
*Jul  2 11:44:42.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/2, changed state to up
*Jul  2 11:44:42.035: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/3, changed state to up
*Jul  2 11:44:42.035: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to up
*Jul  2 11:44:42.035: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed state to down
*Jul  2 11:44:42.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/2, changed state to down
*Jul  2 11:44:42.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/3, changed state to down
*Jul  2 11:44:42.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/0, changed state to down
*Jul  2 11:44:42.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/1, changed state to down
*Jul  2 11:44:42.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/2, changed state to down
*Jul  2 11:44:42.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/3, changed state to down
*Jul  2 11:44:42.375: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 12.4(24)T5, RELEASE SOFTWARE (fc3)
Technical Support: http://www.cisco.com/techsupport
```

Connectivity Tests:

Pinging Ubuntu And Pfsense From R1

```
R1#ping 192.168.50.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/28 ms
R1#ping 192.168.50.21

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.21, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/22/40 ms
R1#
```

Company A To Comapny B

```
PC1> ping 192.168.20.100
192.168.20.100 icmp_seq=1 timeout
192.168.20.100 icmp_seq=2 timeout
84 bytes from 192.168.20.100 icmp_seq=3 ttl=60 time=122.972 ms
84 bytes from 192.168.20.100 icmp_seq=4 ttl=60 time=124.220 ms
84 bytes from 192.168.20.100 icmp_seq=5 ttl=60 time=122.105 ms

PC1>
```

Client PC To Server And PFsense:

```
PC2> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=61 time=91.901 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=61 time=92.333 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=61 time=91.665 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=61 time=91.293 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=61 time=91.251 ms

PC2> ping 192.168.50.21
84 bytes from 192.168.50.21 icmp_seq=1 ttl=60 time=92.713 ms
84 bytes from 192.168.50.21 icmp_seq=2 ttl=61 time=91.294 ms
84 bytes from 192.168.50.21 icmp_seq=3 ttl=61 time=92.834 ms
84 bytes from 192.168.50.21 icmp_seq=4 ttl=61 time=92.996 ms
84 bytes from 192.168.50.21 icmp_seq=5 ttl=61 time=91.406 ms
```

Challenges Faced:

Issues encountered included Cloud connectivity, Layer-2 communication, VMware networking, routing verification, pfSense routing, and Ubuntu service configuration. These were resolved through routing verification, ARP inspection, packet captures, firewall rule validation and service troubleshooting.

Conclusion:

The project successfully demonstrates an enterprise ISP infrastructure integrating Cisco routing, firewall security and Linux server services. OSPF and eBGP provide dynamic routing, pfSense secures the network, and Ubuntu delivers centralized services including Apache, DNS, SSH and Syslog.

References:

- Cisco IOS Documentation
- RFC 2328 (OSPF)
- RFC 4271 (BGP-4)
- pfSense Documentation
- Ubuntu Server Documentation
- Apache HTTP Server Documentation
- BIND9 Administrator Guide

